

BINTULU, Malaysia

WMO: 964410

Lat: 3.124N

Lon: 113.020E

Elev: 74

StdP: 14.66

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	72.0	73.0	69.8	109.7	77.7	70.9	114.4	76.9	13.1	82.8	11.9	83.4	3.9	100	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.6	92.8	80.8	91.6	80.5	90.9	80.3	82.8	89.8	82.0	88.8	81.5	88.2	6.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.8	160.6	87.9	80.3	157.7	87.2	79.2	152.1	85.5	46.9	90.2	46.0	88.9	45.3	88.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	10.3	9.4	70.6	95.5	0.9	2.1	70.0	97.0	69.5	98.2	69.0	99.4	68.4	100.9		
			69.5	83.9	0.8	1.5	68.9	85.0	68.4	85.8	67.9	86.7	67.3	87.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	80.3	80.5	81.4	82.1	82.7	82.2	82.1	82.0	81.6	81.2	81.1	80.8	
	DBStd	1.83	1.68	1.55	1.51	1.58	1.76	1.82	1.88	2.00	1.76	1.66	1.37	1.48	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	11503	939	853	974	964	1013	965	995	992	949	969	934	956	
	CDD65	6028	474	433	509	514	548	515	530	527	499	504	484	491	
	CDH74	56953	4022	3798	4837	5147	5679	5109	5209	5134	4721	4605	4384	4308	
	CDH80	19728	1125	1051	1600	1875	2184	1918	2002	1948	1707	1570	1418	1330	
Wind	WSAvg	3.9	4.0	4.1	4.1	3.9	3.8	3.8	3.8	3.9	4.0	4.0	3.8	4.0	
Precipitation	PrecAvg	149.8	17.8	11.0	10.9	10.5	9.0	9.9	9.8	11.8	11.4	13.9	16.6	17.6	
	PrecMax	212.0	50.4	31.6	21.9	22.4	18.6	24.1	23.1	26.7	21.7	30.2	25.9	30.7	
	PrecMin	107.2	3.1	3.0	1.3	2.4	2.3	0.0	2.8	2.2	4.0	5.4	4.6	9.1	
	PrecStd	20.9	9.6	6.6	5.6	4.4	3.8	4.5	4.4	5.9	4.4	4.8	4.6	4.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	89.8	91.4	92.5	93.4	93.4	93.3	93.5	92.8	91.6	91.1	90.2	
		MCWB	80.4	80.3	80.4	81.4	81.2	81.3	80.4	79.9	80.6	80.3	80.6	80.6	
	2%	DB	88.1	88.1	89.7	91.1	91.7	91.6	91.5	91.7	91.1	89.9	89.5	89.2	
		MCWB	79.9	79.7	79.8	81.0	80.8	80.7	80.1	80.2	80.3	79.8	80.0	80.1	
	5%	DB	86.8	87.0	88.2	89.6	90.6	90.0	90.0	90.0	89.6	89.0	88.0	87.8	
		MCWB	79.1	79.2	79.6	80.1	80.4	80.0	79.7	79.7	79.6	79.5	79.5	79.6	
	10%	DB	85.6	85.7	87.3	88.1	89.3	88.8	88.9	88.6	88.0	87.5	86.7	86.3	
		MCWB	78.5	78.6	79.3	79.7	80.0	79.6	79.4	79.4	79.4	79.2	79.1	78.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.8	81.7	82.7	83.4	83.2	83.5	83.0	83.2	83.0	81.9	82.3	82.0
			MCDB	88.1	87.6	88.4	89.6	90.8	91.3	90.7	90.5	90.1	89.3	88.3	88.4
2%		WB	80.6	80.5	81.1	81.8	81.9	82.0	81.6	81.7	81.6	80.9	81.0	80.9	
		MCDB	86.5	86.4	87.3	88.6	89.1	89.0	88.9	88.8	88.4	87.7	87.0	87.0	
5%		WB	79.7	79.6	80.2	81.0	81.2	81.0	80.8	80.7	80.5	80.0	80.1	80.0	
		MCDB	85.1	85.0	86.3	87.4	88.2	87.8	87.9	87.7	87.2	86.4	85.9	85.8	
10%		WB	79.0	79.0	79.5	80.1	80.5	80.0	79.9	79.8	79.7	79.3	79.4	79.2	
		MCDB	83.9	84.0	85.2	86.1	87.1	86.5	86.5	86.3	85.9	85.4	84.9	84.5	
Mean Daily Temperature Range	5% DB	MDBR	11.5	11.4	12.5	13.2	13.6	13.8	14.1	14.1	13.6	13.1	12.7	12.2	
		MCDBR	13.3	13.1	14.1	14.9	15.6	15.8	15.7	16.1	15.8	15.1	14.2	13.9	
		MCWBR	6.0	5.9	6.2	6.4	6.5	6.9	6.6	6.9	7.1	6.6	6.4	6.3	
	5% WB	MCDBR	12.5	12.2	13.2	14.0	14.4	14.9	14.9	15.2	14.8	14.0	13.4	13.1	
		MCWBR	6.0	5.9	6.3	6.5	6.6	6.6	7.2	7.0	7.2	6.6	6.5	6.4	
Clear-Sky Solar Irradiance	taub	0.416	0.429	0.430	0.412	0.408	0.412	0.417	0.478	0.483	0.484	0.414	0.407		
	taud	2.470	2.418	2.412	2.462	2.482	2.477	2.455	2.263	2.248	2.249	2.465	2.501		
	Ebn at Noon	288	286	284	284	278	273	273	262	266	267	286	289		
	Edh at Noon	36	39	39	36	35	34	35	44	45	45	36	35		
All-Sky Solar Radiation	RadAvg	1435	1568	1694	1710	1614	1563	1599	1618	1561	1496	1492	1456		
	RadStd	112	96	75	66	63	74	70	62	100	108	62	75		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.06	+0.85	+0.52	N/A	N/A	+202	+205
Regional (2 neighbors)	N/A	N/A	N/A	+0.58	+0.47	+0.38	N/A	N/A	+119	+119

Nomenclature: See separate page